

Stabilizing and Accelerating Autofocus with Expert Trajectory Regularized Deep Reinforcement Learning

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Abstract

Autofocus is a crucial component of modern digital cameras. While recent learning-based methods achieve state-of-the-art in focus prediction accuracy, they unfortunately ignore the potential focus hunting phenomenon of back-and-forth lens movement in the multi-step focusing procedure. To address this, in this paper, we propose an expert regularized deep reinforcement learning (DRL)-based approach for autofocus, which can utilize the sequential information of lens movement trajectory to both enhance the multi-step in-focus prediction accuracy and reduce the chance of focus hunting. Our method generally follows an actor-critic framework. To accelerate the DRL's training with a higher sample efficiency, we initialize the policy with a pre-trained single-step prediction network, where the network is further improved by modifying the output of absolute in-focus position distribution to the relative lens movement distribution to establish a better mapping between input images and lens movement. To further stabilize DRL's training with a lower occurrence of focus hunting in the resulting lens movement trajectory, we generate some offline trajectories based on prior knowledge to avoid focus hunting, which are then leveraged as an offline dataset of expert trajectories to regularize the actor network's training. Empirical evaluations show that our method outperforms those learning-based methods on public benchmarks, with higher single- and multi-step prediction accuracy, and a significant reduction of focus hunting rate.

1. Introduction

Autofocus (AF) is a process through which the camera can determine automatically a lens position to focus on the region of interest (RoI). It is a crucial step in image capturing, since once an image is already captured in a defocused and blurry state, it is then difficult to restore its sharpness

through any post-processing due to the varying effects of blur influenced by factors such as camera aperture and radial distortion [1]. In addition, the beauty of a moment often lies in its fleeting nature, which thus imposes an imperative demand for the fast focusing speed of an AF algorithm. However, due to the instability in focusing systems incurred by some interference factors, the AF mechanism of the camera lens may fail to accurately lock onto the focal point, resulting in a repeated movement of the lens within a certain range. This phenomenon, as illustrated in Fig. 1, is referred to as *focus hunting (FH)*, whereby the camera is unable to identify the focus plane and thus drives the lens forward and backward through a set range of focus hunting for it. It then causes variations in the image sharpness and changes the lens's field of view (FoV), ultimately leading to an unstable focusing experience with longer focusing time, which is a common issue in the traditional AF-assisted cameras [2]. It is thus of great importance to develop an AF algorithm that not only achieves a higher in-focus accuracy but can also stabilize the trajectory of lens movement with a faster convergence speed to the in-focus position, thus reducing the probability of occurrence of focus hunting [43, 48].

There is a large body of work on the traditional autofocus methods, generally falling into two categories: search-based and phase-based methods. Search-based AF methods [40, 42] search within a range of focus distances to locate the lens position of the extreme sharpness value, which is then determined as the in-focus lens position. Thus, they need to move the lens back and forth, which leads to a highly accurate identification of the in-focus position, but with an annoyingly slow focusing procedure and inevitably the occurrence of focus hunting. Phase-based approaches, on the other hand, rely on the dual-pixel (DP) sensors to obtain the phase data from the left and right sub-images, which is then leveraged to predict the lens movement and determine the in-focus position in a single step of lens

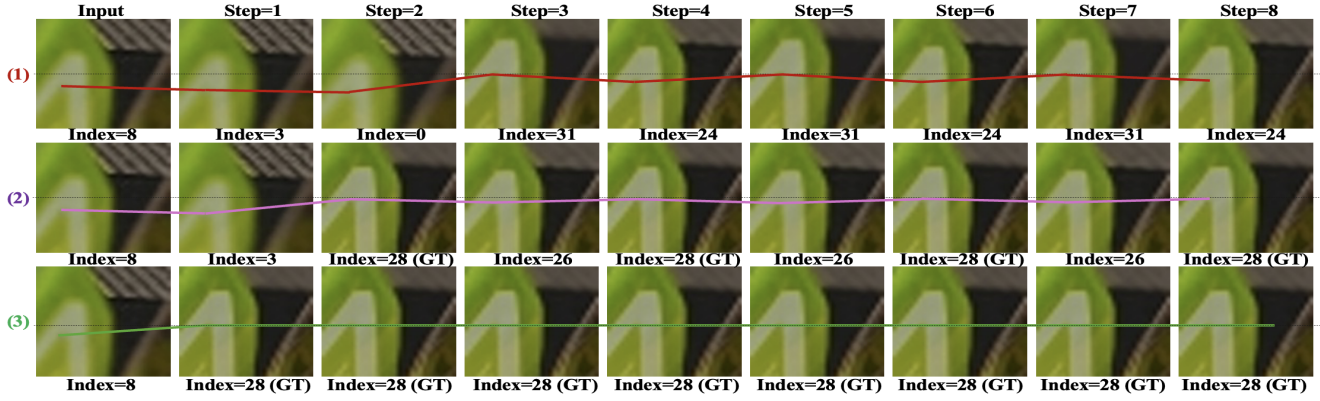


Figure 1. Visualization of the focus hunting phenomenon. The lens moves back and forth within a certain range, resulting in a slight change of field of view (FoV) and sharpness. The red, purple, and green lines mark the trajectory of the same target’s locations in the picture obtained by three models. The yellow line is a horizontal line as a location reference. (1) The trajectory generated by the learning-based method [7] that predicts the lens position directly. (2) The trajectory generated by the learning-based method [7] that predicts the relative movement of lens. (3) The trajectory generated by our DRL-based method. It can be seen that the FOV and sharpness in trajectory (1) change more significantly, while those in trajectory (2) change slightly, and the lens terminates at the GT after Step 2 in trajectory (3).

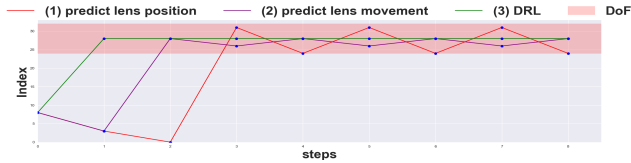


Figure 2. Trajectories generated by predicting the lens position [7] (red line), predicting the relative movement (purple line), and DRL (green line). The trajectory generated by our DRL is the smoothest, indicating a stable and fast focusing procedure.

movement. Though accelerating the focusing procedure, the phase data obtained by sensors is highly susceptible to noise, leading to a relatively low in-focus accuracy. Especially in some extreme cases, e.g., the low-light conditions, the phase detection may need to be executed multiple times at different lens positions to predict a more accurate in-focus position, still at the risk of focus hunting.

Recent attempts [7, 16] have concentrated on addressing the above trade-off between accuracy and speed, by exploiting the expressive power of neural networks to directly learn a mapping from the captured image at any lens position to the predicted in-focus position. However, when we directly treat the learned neural network as a lens movement policy and apply it at multiple steps to adjust the lens movement, we notice that some of the resulting trajectories still suffer from the back-and-forth movement of the lens, as shown in Fig. 1 and Fig. 2, due to the lack of leveraging the sequential information obtained from multiple exploration steps and the ignorance of considering future effects of the current prediction. Deep reinforcement learning (DRL), with successful applications in sequential decision-making tasks such as autonomous driving [12] and robot control [8], has enabled the agent to adaptively adjust its policy through iterative interactions with the environment, based both on the immediate reward of current decision step and the long-term reward over the entire sequence of decisions.

Inspired by this, we propose in this paper an expert regularized DRL-based approach for camera autofocus, which is able to utilize the sequential information of lens movement trajectory to both enhance the multi-step in-focus prediction accuracy and reduce the chance of focus hunting. Our proposed method generally follows an actor-critic framework, within which the actor network (i.e., policy network) takes the captured left and right dual-pixel images with the positional information of RoI and lens as the state input, and sequentially generates a trajectory by making decisions on the lens movement. The critic network, on the other hand, receives the same input as the actor network and estimates the future expected accumulated reward given the current state, which is used to evaluate the action predicted by the actor network and thus eliminate suboptimal actions. To accelerate the DRL’s training with a higher sample efficiency, we initialize the actor network with a pre-trained single-step prediction network based on the MobileNet-V2 architecture [33], where to be more consistent with the environment transition in AF tasks, the output of absolute in-focus position distribution as used in [7] is modified to the relative lens movement distribution. The critic network also takes the same MobileNet-V2 architecture. To further stabilize the DRL’s training with lower occurrence of focus hunting in the resulting lens movement trajectory, we generate some offline trajectories based on prior knowledge to avoid focus hunting, which are then leveraged as an offline dataset of expert trajectories to regularize the actor network’s training. Our main contributions are summarized as follows:

- To overcome the limitation of single-step prediction networks, we formulate AF as a multi-step task and then propose a DRL-based framework to fully utilize the sequential information of lens movement trajectory and adjust the policy based on the immediate and long-term rewards.
- We initialize the policy with a pre-trained single-step pre-

diction network, where the network is improved by modifying the output of absolute in-focus position distribution to relative lens movement distribution to establish a better mapping between input images and lens movement.

- We construct an offline dataset of expert trajectories based on prior knowledge, regularizing the training of the policy network towards reducing occurrence of focus hunting.
- Empirical evaluations demonstrate that our method outperforms state-of-the-art AF methods on public benchmarks, with higher single- and multi-step prediction accuracy, and a significant reduction of focus hunting rate.

2. Related Works

Traditional passive AF methods can generally be categorized into search-based and phase-based approaches. Search-based approaches calculate and maximize the sharpness of the images based on image statistics [9, 21, 40, 42, 53] or frequency-domain image representations [18, 24, 25, 47, 50] to determine the optimal lens position. While showing high accuracy, these approaches are time-consuming. Additionally, search-based methods inevitably lead to focus hunting, since the peak must be surpassed to locate the exact focus point. Though some methods [14, 39] can reduce the search steps, they cannot solve the problem of focus hunting completely, or can only work on some specific devices.

With the development of dual-pixel sensors, phase-based methods have become the primary AF method in modern digital cameras [37]. With the phase information of the two (left/right) split pixels, the disparity can be calculated to determine the lens movement. Consequently, phase-based approaches can move the lens to the in-focus lens position with a single image capture, which is more efficient than search-based approaches. However, factors that are difficult to model, such as lens shading, can affect the calculation of phase, leading to lower accuracy. Numerous works adopt learning-based methods to address the trade-off between accuracy and speed. But these works either apply to specific tasks like microscopy images [31, 51], or are based on phase information obtained by specific hardware [17, 46], which are difficult to reproduce and compare.

The above methods concentrate on the processed sharpness metrics or phase data, not sufficiently using the information from pictures. Inspired by the stereo vision [19, 20], information obtained from dual-pixel pictures has found increasingly widespread applications in the computer vision community, such as monocular depth estimation [11, 22] and defocus deblurring [1]. In AF, Herrmann *et al.* [16] utilize convolutional networks to predict the in-focus position from input images and provide a large-scale AF dataset. Choi *et al.* [7] improve Herrmann's network by incorporating the region of interest position encoding (RoI-PE) and the lens position encoding (Lens-PE), significantly reducing the size while achieving state-of-the-art performance.

Both methods, similar to the phase-based approaches, essentially employ the DP data to estimate the disparity and predict the lens position. But the difference is that phase-based approaches estimate the lens movement through disparity rather than directly predicting the next lens position. Meanwhile, though [7, 16] discuss the multi-step prediction, they only address the accuracy of multi-step prediction without considering the potential for focus hunting, which can greatly affect the speed and accuracy of the AF model.

The main reason for focus hunting in learning-based methods is the lack of consideration of the sequential information and the future effects of the actions. Yan *et al.* [49] and Wang *et al.* [46] adopt the long short-term memory (LSTM) to enhance the model's ability to incorporate previous information. With extra computation, the ability of LSTM to capture the 2-dimension data is still limited. The paradigm of predicting actions from states and the need to incorporate sequential information and future effects inspire us to use reinforcement learning (RL) in AF. Though Yu *et al.* [54] and Chan *et al.* [6] have proposed DRL-based autofocus methods, they both concentrate on the design of reward functions and do not fully leverage the advantages of RL. Wang *et al.* [46] adopt RL in their work to dynamically choose the RoI, rather than predicting the lens position or movement. To the best of our knowledge, little research has been conducted to reduce the occurrence of focus hunting while accelerating the multi-step prediction with a higher accuracy for AF tasks.

3. Proposed Method

3.1. Problem Formulation

We formulate the problem of AF as presented in [7], where the candidate lens position is discretized into n focal indices. Given an input image I captured at a lens position $k \in \{1, 2, \dots, n\}$, we extract a patch I_k^p corresponding to the region of interest (RoI) that we aim to focus, where $p \in \{1, 2, \dots, P\}$ denotes the spatial position of the RoI in this image. A qualified AF process should be able to predict the correct focal index of the lens from a patch sequence captured by moving the lens, while the sequence should ideally be as short as possible, and without a back-and-forth movement. Considering that modern digital cameras (e.g., in smartphones) are commonly equipped with the advanced dual-pixel imaging system that can provide the phase information for AF, we specifically pay attention to the dual-pixel data, such that the input image I consists of two channels. Thus, the extracted patch at the k -th lens position, with resolution $h \times w$, is denoted as $I_k^p \in \mathbb{R}^{2 \times h \times w}$. The terms *focal stack* and *single slice* are defined as the set of patches $\{I_k^p\}_{k=1}^n$ extracted from all lens positions and a single patch I_k^p , respectively. The objective is to predict the in-focus lens position k^* , which is assumed to be unique for

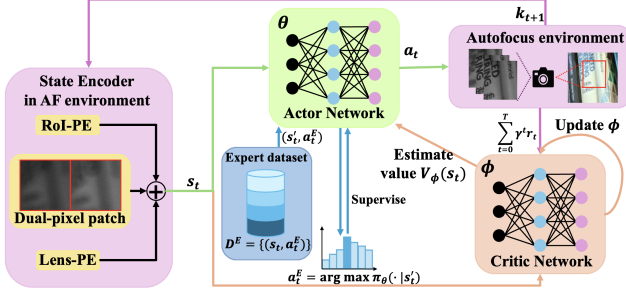


Figure 3. Overall framework of our DRL-based AF model with expert trajectory regularization. The green, orange, blue, and purple boxes specify the actor network, critic network, expert dataset, and environment, respectively.

each focal stack.

Relying on the different types of patch sequences used for AF, our AF problem falls into the multi-step prediction category [7], which can be formulated as:

$$I_{k_0}^p \mapsto k_1, I_{k_1}^p \mapsto k_2, \dots, I_{k_t}^p \mapsto k^*. \quad (1)$$

Starting from a random focal index k_0 , it makes a prediction step-by-step, generating a trajectory of the lens positions $\{k_0, k_1, k_2, \dots, k^*\}$. Instead of directly predicting the absolute lens position as in Eq. (1), in this paper, we consider predicting the relative lens movement at each step, and thus reformulate the multi-step setting of the AF problem as:

$$I_{k_0}^p \mapsto k_1 = k_0 + a_0, \dots, I_{k_t}^p \mapsto k^* = k_t + a_t, \quad (2)$$

where a_t is the relative movement of the lens at time step t .

3.2. Overall Framework of DRL-Based AF

The framework of our DRL-based AF method is shown in Fig. 3, with three major components: an *actor network*, a *critic network*, and an *offline dataset of expert trajectories*.

- *Actor network (i.e., policy network) parameterized by θ* : At each time step t , it takes the state s_t , including the left and right DP images, the RoI-PE and the Lens-PE, as the input, and outputs an action $a_t \sim \pi_\theta(\cdot|s_t)$, where the sign of a_t indicates the direction of the lens movement and the absolute value $|a_t|$ represents the moving distance.
- *Critic network parameterized by ϕ* : At each time step t , it takes the state s_t as input, and outputs the estimated value function $V_\phi(s_t)$ for the current state s_t . This value function represents the future expected accumulated reward given the current state, which thus can be used to evaluate the action predicted by the actor network, allowing the DRL agent to identify and eliminate some suboptimal actions. Since this evaluation is based on the future expected accumulated reward, the actions are thus chosen based on their anticipated long-term impact, rather than the immediate return.
- *Offline dataset of expert trajectories $\mathcal{D}^E = \{(s_t, a_t^E)\}$* : It provides the expert trajectories with state s_t and as-

sociated expert action a_t^E . These trajectories are generated based on the prior knowledge to avoid focus hunting, and can thus be used to regularize the training of the actor network by maximizing the probability of expert action given a state s_t , which ensures that the actor network learns from the high-quality multi-step movement of the lens from the expert.

3.3. Actor Network for Movement Prediction

To utilize the sequential information of the lens movement trajectory to both enhance the multi-step in-focus prediction accuracy and reduce the chance of focus hunting, the actor network ideally should establish a more accurate mapping between the disparity from DP data and the lens movement.

The AF procedure can be formulated as a Markov Decision Progress (MDP), defined by a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma \rangle$, where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ denotes the action space and $\gamma \in (0, 1]$ is the discount factor. At each time step t , a state $s_t \in \mathcal{S}$, consisting of the left and right DP images, the RoI-PE and the Lens-PE, is received by the RL agent, and an action $a_t \in \mathcal{A}$ is then generated by the actor network to determine the lens movement. The environment transition dynamics $P(s_{t+1}|s_t, a_t)$ in AF can be defined as:

$$s_{t+1} = g(\text{clip}(a_t + g^{-1}(s_t), 0, k_{max})), \quad (3)$$

where $g(k_t) = s_t$ maps the lens position k_t to the state s_t at time-step t and $k_{max} \triangleq k_n$ is the maximum lens position. The reward function $R(s_t, a_t)$ is defined as the negative absolute distance between the predicted and ground-truth in-focus positions, where we additionally introduce a penalty term R_{fh} to mitigate the occurrence of focus hunting:

$$R(s_t, a_t) = -|k_{t+1} - k_{gt}| + R_{fh}, \quad (4)$$

where k_{gt} denotes the ground-truth lens position, and the penalty term R_{fh} is defined as:

$$R_{fh} = \begin{cases} c, & \text{if FH occurs;} \\ 0, & \text{otherwise;} \end{cases} \quad (5)$$

where c is a negative hyperparameter that penalizes the event of focus hunting. The main goal of RL is to maximize the expected return $J(\theta)$ defined as:

$$J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right], \quad (6)$$

where π_θ is the policy network parameterized by θ and T denotes the maximum time step.

Since the single-step prediction network model proposed by Choi *et al.* [7] has demonstrated a state-of-the-art accuracy, we adopt its network architecture as our actor network.

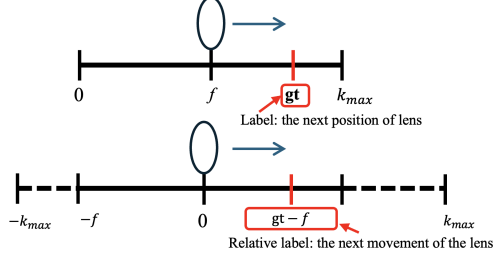


Figure 4. Illustration of relative movement label.

However, the original model in [7] directly predicts the in-focus position, which contradicts the principle of using DP data for in-focus prediction. In essence, since there is a clear correspondence between the disparity and lens movement (i.e., through the defocus conversion coefficient), it is more reasonable to predict the lens movement rather than the lens position based on the DP data. Thus, we modify the original network’s output of absolute in-focus position distribution in [7] to the relative lens movement distribution, with the inference procedure of actor network shown in Fig. 9. Accordingly, Fig. 4 illustrates the conversion from original absolute position labels to relative movement labels that are used for training. Unlike [17], we expand the action range from $[-gt, k_{max} - gt]$ to $[-k_{max}, k_{max}]$, and the label denoting the relative movement towards GT index then converts to $gt - f$. If the actor network predicts an out-of-bound action, it will be clipped to the closest lower/upper bounds. Based on this formulation, we initialize the actor network with a pre-trained MobileNet-V2 [33] optimized based on the ordinal regression loss [10]. For details about the pre-trained model of actor network, please refer to Sec. 7 in our supplementary.

3.4. Critic Network Incorporating Future Impact

The actor network cannot directly incorporate the future reward of current actions, where the probability of any output action with a larger instant reward will be selected, leading to a suboptimal policy. To address this issue, a critic network is further introduced to provide baselines and evaluate the future impact of the current actions.

The critic network estimates the value function $V_\phi(s_t)$ of the current state s_t , which is defined as the expected future accumulated discount reward of actions. If we further define the future accumulated discount reward of an action under a given state as a Q-function $Q(s_t, a_t)$, the advantage function $\hat{A}_t = Q(s_t, a_t) - V_\phi(s_t)$ then evaluates the relative quality of a particular action under the current state. With this advantage function, the actor network can be trained by minimizing the clipped surrogate objective loss, given by:

$$\mathcal{L}_{clip}(\theta) = \mathbb{E}[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)], \quad (7)$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_\theta^{old}(a_t|s_t)$ is the probability ratio and ϵ is a clip hyperparameter.

Algorithm 1 Expert trajectory generator.

Input: $\mathcal{D} = \{f, GT, (s_0, a_0), \dots, (s_n, a_n)\}_N$;

Output: $\mathcal{D}^E$;

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1:  $\mathcal{D}^E = \emptyset$ ;
2: for all  $\{g(\cdot), GT, (s_0, a_0), \dots, (s_n, a_n)\}_i \in \mathcal{D}$  do
3:   for all  $j = 0, 1, 2, \dots, n$  do
4:      $k_j \leftarrow g_i^{-1}(s_j)$ ;
5:      $o_j \leftarrow \text{sign}(k_0 - GT_i) \cdot |k_j - GT_i| + GT_i$ ;
6:      $o_j \leftarrow \text{CLIP}(o_j, k_0, GT_i)$ ;
7:   end for
8:    $(o_{m_0}, \dots, o_{m_n}) \leftarrow \text{sort}(o_0, \dots, o_n; \text{reverse} = \text{Bool}(k_0 > GT_i))$ ;
9:   for all  $j = 1, 2, \dots, n$  do
10:     $s_j \leftarrow g_i(o_{m_j})$ ;
11:     $a_{j-1} \leftarrow o_{m_j} - o_{m_{j-1}}$ ;
12:   end for
13:    $a_n \leftarrow 0$ ;
14:    $\mathcal{D}^E \leftarrow \mathcal{D}^E \cup \{g(\cdot), GT, (s_0, a_0), \dots, (s_n, a_n)\}_i$ ;
15: end for
16: return  $\mathcal{D}^E$ .
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The critic network, on the other hand, is trained by minimizing the ℓ_2 loss between the estimated value function $V_\phi(s_t)$ and the true value function $\hat{V}_t$:

$$\mathcal{L}_{critic} = \frac{1}{N} \sum_{t=1}^N (V_\phi(s_t) - \hat{V}_t)^2. \quad (8)$$

3.5. Expert Trajectory Regularization

Though the incorporation of a critic network can improve the effectiveness of actor network’s training, the exponential growth in the state and action combinations inherently leads to another issue of low sample efficiency in DRL. Recent literature has witnessed the success of incorporating offline data for improving the sample efficiency of DRL’s training, both theoretically [38, 45] and empirically [5, 27, 28]. Motivated by this, and to further stabilize the DRL with a lower occurrence of focus hunting in the resulting lens movement trajectory, we generate some offline trajectories based on the prior knowledge to avoid focus hunting, which are then leveraged as an offline dataset $\mathcal{D}^E = \{(s_t, a_t^E)\}$ of expert trajectories to regularize the actor network’s training. Algorithm 1 shows our proposed method to generate these expert trajectories, with more details provided in Sec. 8 in our supplementary.

Fig. 5 shows our proposed framework for expert trajectory regularized DRL’s training. At each time, we sample N trajectories from the offline expert dataset $\mathcal{D}^E$ to calculate the imitation loss:

$$\mathcal{L}_{imit}(\theta) = -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \log p(\pi_\theta(\cdot|s_{i,t}) = a_{i,t}), \quad (9)$$

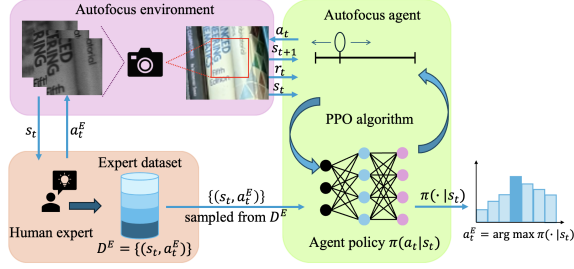


Figure 5. Expert trajectory regularized DRL’s training.

where $a_{i,t} = k_{i,t+1} - k_{i,t}$ denotes the action at time step t in the i -th trajectory, $s_{i,t}$ represents the network input at lens location $k_{i,t}$, and $\pi_\theta(\cdot|s_{i,t})$ represents the output distribution of the actor network, with the total loss defined as:

$$\mathcal{L}(\theta) = \mathcal{L}_{clip}(\theta) + \lambda \cdot \mathcal{L}_{imit}(\theta), \quad (10)$$

where λ balances the online and offline learning. By minimizing the loss $\mathcal{L}(\theta)$, our DRL algorithm maximizes the expected accumulated discount reward $J(\theta)$ indirectly.

4. Experiments

In the experiments, our actor and critic networks are built upon MobileNetV2 [33]. We reproduce the models for [16] and [7], since no open-source code is available. In the multi-step prediction setting, we only consider the model obtained by [7], since the multi-step model given by [16] must be re-trained for each step, which is impractical. More results and analysis are provided in Sec. 9 in our supplementary.

Dataset. We adopt the publicly available AF dataset constructed by Herrmann *et al.* [16]. This dataset comprises 51 scenes, each containing 10 stacks with varying compositions. All instances within the dataset are captured at $n = 49$ focal depths, uniformly sampled in the inverse depth space, ranging from 0.102 to 3.91 meters, captured using the Google Pixel 3 smartphones. To create the training and test sets, we extract and sample 128×128 patches with a stride of 96 from the central camera, resulting in a total of 3,341,163 patches for the training set and 382,445 patches for the test set. Patches with the same spatial position within one image and 49 different focal depths constitute a single focal stack. Please refer to [16] for additional details about the dataset, and refer to Sec. 6 in our supplementary for the labeling method adopted in this paper.

Evaluation metric. We follow the evaluation metric proposed by Scharstein *et al.* [34], which is used for Middlebury stereo dataset and adopted by Herrmann *et al.* [16] and Choi *et al.* [7] to evaluate the AF model’s prediction accuracy. In particular, this metric is calculated by the proportion of patches whose stop positions are within 0, 1, 2, and 4 indices away from the in-focus index, which can be considered in the depth of field (DoF). The mean absolute error (MAE) and root mean squared error (RMSE) are also

used to evaluate the AF model’s performance. To evaluate the focus hunting (FH), we retain every focusing trajectory and calculate the direction and distance of each lens movement. If the direction of a current movement is opposite to that of the previous one in the trajectory, we consider an occurrence of FH in that trajectory. Since the actor network obtained by our DRL and the model in [7] share the same architecture and similar input/output, we can simply regard the number of steps to reach the in-focus position as the focus speed. We calculate the images captured at all initial positions of the lens to account for accuracy and FH rate.

Implementation detail. We use PyTorch [29] and TorchRL [4] for our implementation. For deep learning models, we use the Adam optimizer [23] with a learning rate of 10^{-3} , $(\beta_1, \beta_2) = (0.5, 0.999)$, a batch size of 128, and cosine learning rate decay [26] for 10k iterations. For reinforcement learning models, we adopt the PPO-CLIP [35] algorithm. The initial learning rate is set to 1×10^{-5} , and the clip epsilon is 0.2, a batch size of 32, Adam optimizer with the same (β_1, β_2) as deep learning. The maximum episode length of RL is set to 4. For the expert trajectory, we first construct an expert dataset containing 6,000,000 expert trajectories and set the weight λ to 0.001 in the loss function.

4.1. Quantitative Results

FH rate and multi-step prediction accuracy (Effects of DRL). The FH rate and multi-step prediction accuracy of comparison methods are shown in Tab. 1. Though the relative movement label can significantly improve the one-step prediction accuracy, the FH rate in the multi-step setting is still high since the current deep-learning training process cannot obtain and exploit the sequential information of the lens movement. Due to DRL’s advantages in modeling sequential decision-making problems, our DRL-based models can directly decrease the FH rate without relying on any extra networks, such as the LSTM [36]. Compared with the policy obtained by deep learning, our DRL-based policy can decrease the FH rate by 9.1% in 2-step prediction, 16.5% in 3-step prediction, and 18.0% in 4-step prediction, while adopting the same network architecture. Our work shows some advantages over existing RL baselines. Since few features are extracted from the scalar input of [6], it can only guide the lens towards one side with the higher label frequency, causing extremely low accuracy but 0 FH rate. Since [54] generalizes poorly in various views (as stated by the authors), while the dataset in our work contains images with multiple views and object distances, it hardly gives a move signal. The lens thus stops at the start position in most cases, causing small changes in accuracy and a low FH rate.

For the multi-step prediction performance, the accuracy of the baseline [7] increases a little and even decreases in 4-step prediction. The accuracy of the model obtained by the relative movement label can achieve substantial gains

Table 1. Prediction accuracy and focus hunting (FH) rate results in multi-step setting.

Algorithm	Step	= 0 $\uparrow$	≤ 1 $\uparrow$	≤ 2 $\uparrow$	≤ 4 $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	FH $\downarrow$
DRL in [6]	1	0.030	0.071	0.117	0.210	14.216	17.908	NA
	2	0.029	0.050	0.082	0.151	17.214	20.827	0.0
	3	0.029	0.050	0.082	0.151	17.214	20.827	0.0
	4	0.029	0.050	0.082	0.151	17.214	20.827	0.0
DRL in [54]	1	0.022	0.062	0.102	0.181	15.575	19.143	NA
	2	0.023	0.064	0.105	0.185	15.446	19.051	0.019
	3	0.024	0.065	0.106	0.187	15.437	19.073	0.025
	4	0.024	0.065	0.106	0.186	15.500	19.142	0.030
RoI-PE & Len-PE [7]	1	0.243	0.584	0.756	0.882	2.839	6.880	NA
	2	0.275	0.640	0.803	0.912	2.454	6.296	0.275
	3	0.279	0.644	0.810	0.914	2.434	6.205	0.362
	4	0.276	0.642	0.807	0.913	2.442	6.277	0.383
[7] with relative label (Ours)	1	0.330	0.698	0.826	0.907	2.342	6.339	NA
	2	0.413	0.795	0.888	0.934	1.894	5.855	0.261
	3	0.427	0.805	0.894	0.937	1.892	5.721	0.340
	4	0.423	0.804	0.893	0.936	1.862	5.844	0.358
Expert Trajectory (Ours)	1	0.336	0.711	0.839	0.915	2.146	5.827	NA
	2	0.439	0.808	0.896	0.941	1.755	5.597	0.170
	3	0.450	0.816	0.900	0.943	1.711	5.532	0.175
	4	0.449	0.818	0.901	0.944	1.694	5.479	0.178

Table 2. Average steps to stop with different termination signals.

Termination Rule	= 0 $\downarrow$	≤ 1 $\downarrow$	≤ 2 $\downarrow$	≤ 4 $\downarrow$	Stop Model $\downarrow$	MAE $\downarrow$	RMSE $\downarrow$
RoI-PE & Lens-PE [7]	6.651	3.644	2.477	1.707	2.208	2.611	6.146
[7] with relative label (Ours)	4.982	2.246	1.704	1.453	1.884	1.906	5.463
Expert Trajectory (Ours)	5.014	2.257	1.683	1.361	1.821	1.874	5.332

of 13.8% prediction accuracy for exact ($= 0$) predictions in 2-step prediction, while our DRL-based model further enhances this accuracy. Furthermore, as the number of lens movement steps increases, the accuracy of deep learning-based methods may decline, whereas the accuracy of our DRL-based method remains unchanged or even improves, demonstrating the stability of our DRL-based model in multi-step prediction. Moreover, Tab. 2 shows that our models take fewer steps to stop at different termination signals, e.g., $= 0$, ≤ 1 , ≤ 2 , and ≤ 4 . Since in-focus positions remain unknown in practical camera deployment, the above rules may not be feasible. Alternatively, we train a stop model to generate the stop signal, which is modeled as a binary classification task and takes the same architecture and input as our policy network. Tab. 2 shows the average steps, MAE, and RMSE of different policies with this model. Our method still achieves a better prediction with fewer steps. In conclusion, results in Tab. 1 and Tab. 2 show that our methods can achieve higher accuracy with fewer steps than that proposed by [7], demonstrating a faster focusing speed.

Single-step accuracy (Effects of relative movement as labels). The comparison results in the single-step prediction setting are shown in Tab. 3. Compared with classic statistics-based or depth estimation methods, the deep-learning based models “Learning to Autofocus [16]” and “RoI-PE & Len-PE [7]” significantly improve the one-step prediction accuracy. By modifying the output of the model in [7] from the absolute lens position to the relative movement, we can further improve the accuracy by 8.7% for exact ($= 0$) prediction and 2.5% for ≤ 4 focal index difference. Moreover, since the dataset has low-contrast images with similar features but distant labels, minimizing the loss makes the policy perform badly. Via optimizing multi-step

Table 3. Single-step prediction results.

Algorithm	= 0 $\uparrow$	≤ 1 $\uparrow$	≤ 2 $\uparrow$	≤ 4 $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$
ZNCC Disparity with Calibration	0.065	0.179	0.268	0.419	9.476	14.211
SSD Disparity [44]	0.101	0.264	0.386	0.525	8.012	12.414
Learning the Depth [11]	0.112	0.297	0.415	0.562	7.419	12.312
Learning to Autofocus [16]	0.168	0.467	0.659	0.851	3.249	7.619
RoI-PE & Lens-PE [7]	0.243	0.584	0.756	0.882	2.839	6.880
[16] with relative label (Ours)	0.181	0.493	0.684	0.861	3.021	7.374
[7] with relative label (Ours)	0.330	0.698	0.826	0.907	2.342	6.339
Expert Trajectory (Ours)	0.336	0.711	0.839	0.915	2.146	5.827

Table 4. Performance comparison of DRL models with and without Expert Regularization.

Algorithm	Step	= 0 $\uparrow$	≤ 1 $\uparrow$	≤ 2 $\uparrow$	≤ 4 $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	FH $\downarrow$
w/o Expert Regularization	1	0.031	0.056	0.086	0.143	22.005	26.048	NA
	2	0.029	0.065	0.095	0.148	23.845	27.975	0.992
	3	0.037	0.085	0.123	0.188	20.957	25.501	0.993
	4	0.031	0.068	0.099	0.152	23.722	27.889	0.993
Expert Trajectory (Ours)	1	0.336	0.711	0.839	0.915	2.146	5.827	NA
	2	0.439	0.808	0.896	0.941	1.755	5.597	0.170
	3	0.450	0.816	0.900	0.943	1.711	5.532	0.175
	4	0.449	0.818	0.901	0.944	1.694	5.479	0.178

behavior, RL can explore extra information and improve 1-step prediction accuracy.

Effects of expert regularization. As shown in Tab. 4, the policy obtained by the vanilla DRL algorithm without expert regularization has a poor performance, since the action space is a discrete number of 97, which is large for DRL’s training. The expert trajectory significantly improves the performance of the policy given by the PPO algorithm.

4.2. Qualitative Results

Fig. 1 qualitatively visualizes the trajectory of patches and lens positions of different comparison methods, including the policy obtained by [7], the policy obtained by relative movement label, and our DRL-based model. Compared with directly predicting the lens position [7], our relative movement label model is faster by predicting the lens movement within DoF in the second step, but the FH issue cannot be overcome. In comparison, our DRL-based model can generate a faster and more stable prediction by predicting the GT location at the first step and stopping at this index. The lens trajectory of the DRL-based model (the green curve in Fig. 2) is slightly smoother and more stable without the occurrence of FH.

In Fig. 6, we show qualitative results of 4-step predictions of the three models. From the first row, we can observe that all methods can predict the index within DoF, but our methods can give a more precise prediction with a sharper image than Choi’s [7]. From the second to fourth rows, we can observe that [7] sometimes breaks and predicts indices with sharp background, while our DRL-based model successfully predicts indices with sharp target objects. This case is a generally challenging scenario in the AF task since many different DoFs can be estimated. These DoFs may confuse the models, leading to a focus on the wrong object. In general, our relative movement label can give a more accurate prediction by mapping the disparity to the movement of the lens, and the DRL-based model can further enhance the accuracy by estimating the future reward of the action

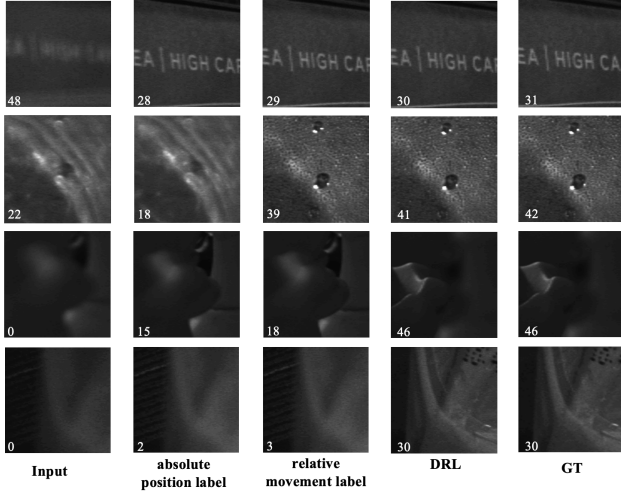


Figure 6. Comparison of the 4-step results of the policy obtained by [7], our relative movement label, and our DRL-based model.

and reducing the focus hunting rate.

In Fig. 7, we analyze our model’s failure cases: (a) it is difficult and meaningless to focus on the images with no texture since the phase information or contrast information is little to estimate the depth, (b) though our models can give a more accurate prediction in scenarios with multiple DoFs, when the target object is small and the background is complex, the models may be confused by these DoFs, leading to focusing on the wrong object, (c) like all DP data-based AF methods, our models may work poorly when the objects are horizontal. Some possible solutions may work to overcome these limitations. For (a), larger patches can provide more phase information, and we also think (a) might not be that important since few people take photos with no texture. For (b), we claim it is not a bad issue, because sometimes people may want to focus on the background obscured by meaningless foreground elements. To solve this problem, larger patches with more information about the target objects may be useful. Some depth estimation methods [22, 52] can also be used to provide more depth information to help distinguish the objects and background. For (c), solving this problem is challenging in algorithms, while little changes in hardware, especially sensors, could easily mitigate this issue [7].

4.3. Ablation Study

Effects of relative label method. Tab. 5 further demonstrates the effect of relative movement as labels. Our DRL-based model can further enhance the accuracy by 0.6% for exact ($= 0$) prediction and 0.8% for ≤ 4 focal index difference, in comparison to the model in [7] modified with the relative movement label. We further compare the two label methods under three different architectures: MobileNetV2 [33], ResNet50 [15], and MnasNet [41]. The relative movement label method outperforms the absolute lens position label for each architecture. For a fairer comparison,

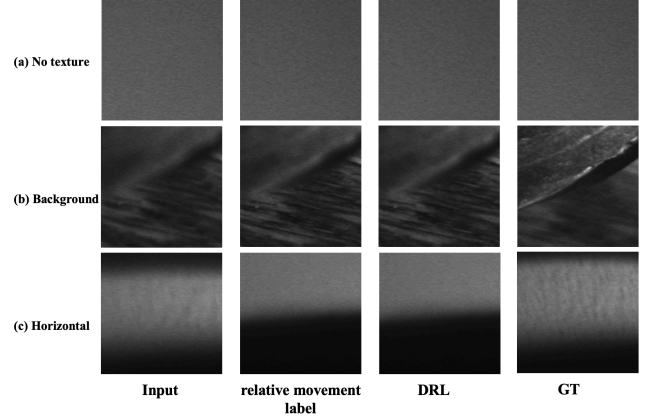


Figure 7. Failure cases of our models, where only the *left* image of DP data is visualized.

Table 5. Single-step prediction results with different network architectures. The number of input channels is 5.

Algorithm	# of params	label	$= 0 \uparrow$	$\leq 1 \uparrow$	$\leq 2 \uparrow$	$\leq 4 \uparrow$	MAE $\downarrow$	RMSE $\downarrow$
MobileNetV2 [33]	2.29M	absolute	0.243	0.584	0.756	0.882	2.839	6.880
	2.35M	absolute	0.252	0.592	0.764	0.888	2.719	6.745
	2.35M	relative	0.330	0.698	0.826	0.907	2.342	6.339
ResNet50 [15]	23.61M	absolute	0.258	0.602	0.789	0.894	2.615	6.623
	23.71M	absolute	0.254	0.596	0.781	0.891	2.655	6.686
	23.71M	relative	0.326	0.709	0.849	0.918	2.136	5.764
MnasNet [41]	1.00M	absolute	0.226	0.568	0.736	0.864	2.983	7.016
	1.06M	absolute	0.213	0.549	0.718	0.850	3.119	7.318
	1.06M	relative	0.305	0.678	0.817	0.905	2.436	6.402

we expand the absolute position label to the same range as the relative movement label. It shows that expanding the number of networks’ output channels can not improve the accuracy, and the effects remain unknown. Since Lens-PE mainly conveys the lens’ current position to the network, while the next position must be predicted by adding its movement to the current position, the relative label methods avoid additionally inferring the identity mapping for Lens-PE, leading to a higher accuracy. To make a fair comparison with the other learning-based models, we thus adopt the same MobileNetV2 architecture in the experiments.

5. Conclusion

We have proposed a DRL-based autofocus model with expert trajectory regularization that can both improve the prediction accuracy and reduce the focus hunting rate, which took an actor-critic framework based on the pre-trained MobileNet-V2 architecture for single-step prediction. For the actor network, we further modified the absolute position label in [7, 16] with the relative movement label method. We also constructed an offline dataset of expert trajectories based on prior knowledge, regularizing the policy network training toward reducing focus hunting. Experimental results have clearly demonstrated the effectiveness of our proposed method, in terms of the multi-step prediction accuracy and the occurrence of focus hunting.

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